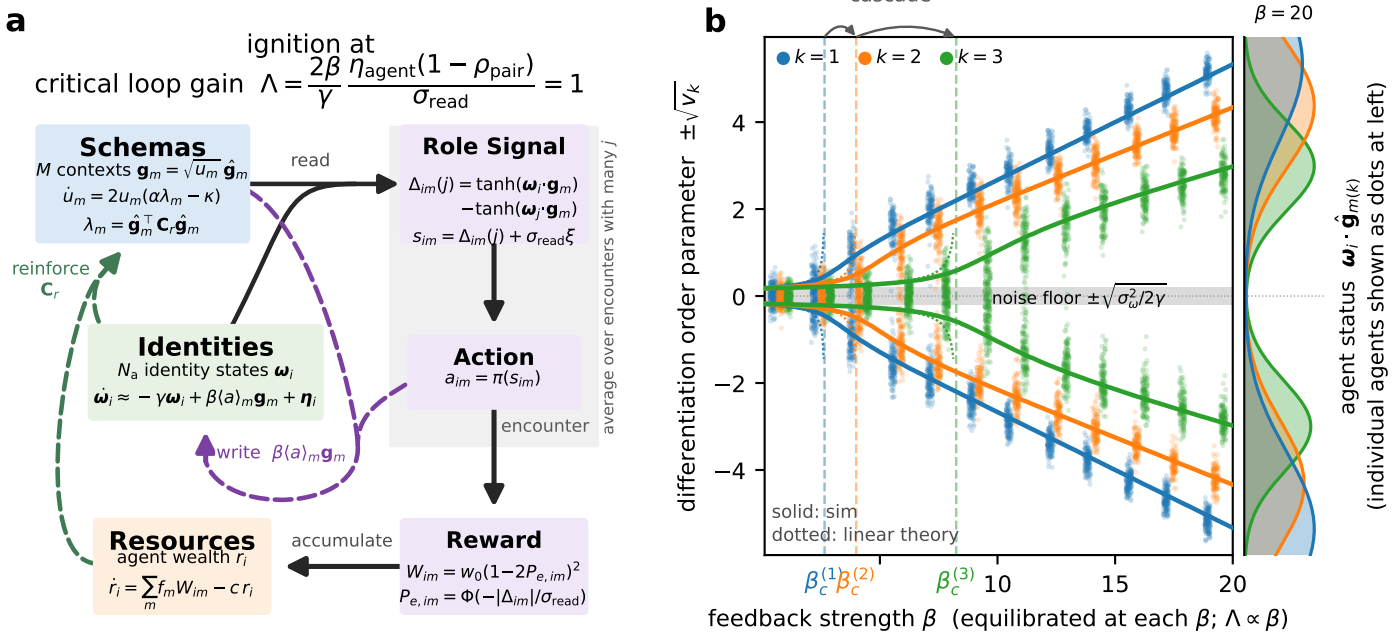

a
ignition at
critical loop gain $\Lambda = \frac{2\beta}{\gamma}\frac{\eta_{\text{agent}}(1-\rho_{\text{pair}})}{\sigma_{\text{read}}} = 1$
Schemas
M contexts $\mathbf{g}_m = \sqrt{u_m}\,\hat{\mathbf{g}}_m$
$\dot{u}_m = 2u_m(\alpha\lambda_m - \kappa)$
$\lambda_m = \hat{\mathbf{g}}_m^\top \mathbf{C}_r \hat{\mathbf{g}}_m$
read
Role Signal
$\Delta_{im}(j) = \tanh(\boldsymbol{\omega}_i\cdot\mathbf{g}_m) - \tanh(\boldsymbol{\omega}_j\cdot\mathbf{g}_m)$
$s_{im} = \Delta_{im}(j) + \sigma_{\text{read}}\xi$
average over encounters with many j
reinforce $\mathbf{C}_r$
Identities
N_a identity states $\boldsymbol{\omega}_i$
$\dot{\boldsymbol{\omega}}_i \approx -\gamma\boldsymbol{\omega}_i + \beta\langle a\rangle_m\mathbf{g}_m + \boldsymbol{\eta}_i$
write $\beta\langle a\rangle_m\mathbf{g}_m$
Action
$a_{im} = \pi(s_{im})$
encounter
Resources
agent wealth r_i
$\dot{r}_i = \sum_m f_m W_{im} - c\,r_i$
accumulate
Reward
$W_{im} = w_0(1-2P_{e,im})^2$
$P_{e,im} = \Phi(-|\Delta_{im}|/\sigma_{\text{read}})$
b
cascade
$\beta = 20$
$k=1$
$k=2$
$k=3$
differentiation order parameter $\pm\sqrt{v_k}$
noise floor $\pm\sqrt{\sigma_\omega^2/2\gamma}$
solid: sim
dotted: linear theory
$\beta_c^{(1)}$ $\beta_c^{(2)}$ $\beta_c^{(3)}$
feedback strength β (equilibrated at each β; $\Lambda \propto \beta$)
agent status $\boldsymbol{\omega}_i\cdot\hat{\mathbf{g}}_{m(k)}$
(individual agents shown as dots at left)